1 • Shared brief and evidence

The spill-mat design lab

Name ____________________________ Date _______________

Fictional design brief and constructed sample data. Compare these particular one-layer 10 cm × 10 cm product squares, not every possible material.

Brief: a removable spill-mat insert must retain at least 6 g of water in every trial. Choose the lowest-cost tested product that meets this. Fictional costs per square: kitchen paper 2 credits; cotton 4 credits; felt 6 credits.

Fixed test method

Dry and tare the weighing tray before each measurement. Weigh a fresh dry sample. Immerse 30 s. Drip vertically 10 s without squeezing. Weigh wet promptly. Retained water = wet − dry mass.

Keep conditions matched

Same area, one layer, water conditions, immersion, timing, weighing tray and scale. Use three fresh dry samples per product.

Plan needing repair

Large felt: soak 60 s and squeeze hard. Small paper: soak 10 s and weigh immediately. No dry mass recorded.

Product	Dry mass each / g	Wet masses T1 / T2 / T3 / g
Kitchen paper	0.8	6.0 / 6.3 / 6.1
Cotton cloth	2.0	9.0 / 8.8 / 9.2
Felt sheet	2.5	10.5 / 10.3 / 10.7

Constructed sample data above: sample thicknesses and structures differ; this compares the named products at equal area, not intrinsic absorption at equal thickness.

W1. Repair at least two parts of the unfair plan. W2. Which products meet the brief? Which is cheapest?

2 • Supported route: fair test to design choice

Use page 1. Read, point, draw or dictate. A calculator is welcome.

S1. What do we deliberately change?

A. Product type

B. Soak time for each product

S2. What do we measure?

A. Retained water mass in g

B. Which colour looks best

S3. Name two things to keep the same. Word bank: sample area / soak time / final result / drip time.

Product	Wet mass T1 / g	Dry mass / g	Retained water / g
Paper	6.0	0.8	
Cotton	9.0	2.0	
Felt	10.5	2.5	

S4. Constructed data from page 1. Complete: wet − dry. Show or dictate one subtraction.

Constructed repeat summary: paper holds 5.2, 5.5, 5.3 g; cotton 7.0, 6.8, 7.2 g; felt 8.0, 7.8, 8.2 g. Every trial must reach 6 g. Costs: paper 2, cotton 4, felt 6 credits.

S5. I choose ____ because all trials reach ____ g or more, and it costs less than ____ .

3 • Standard route: a fair comparison report

Use the constructed measurements on page 1. Your conclusion should answer the product brief.

N1. State the factor you change, the outcome you measure and three conditions you keep the same.

N2. Write the key measurement steps, including the dry start, soak time, drip time and repeats.

Product	Water T1 / g	Water T2 / g	Water T3 / g
Paper			
Cotton			
Felt			

N3. Calculate and record water retained from the constructed wet and dry masses on page 1. Show one subtraction.

3 continued • Standard route: the design decision

Use your calculations and the common evidence page. Explain the decision and the next useful test.

N4. Recommend the cheapest passing product. Quote its lowest trial result and compare the cost with the other passing product.

N5. State one reason this does not prove a material is best for every mat, and one next test.

4 • Stretch route: test the rule as well as the sample

Use the constructed sample data and fictional cost brief on page 1.

T1. Calculate retained water for all nine samples. Calculate the mean for each product, showing one calculation. A calculator is welcome.

T2. Why is the mean alone not enough for our every-trial requirement? Use this extra constructed set: 5.0, 7.0, 9.0 g.

T3. Recommend the lowest-cost passing product from page 1. State one uncertainty and a practical improvement.

T4. Our squares have equal area but different dry masses and thicknesses. What question did our test answer? How would a comparison of water retained per gram of product answer a different question?

T5. Propose a second measurable property for a reusable mat. Give a fair comparison method with two controls.

5 • Exit and optional practical record

Choose the correct label: paper evidence route / actual practical results. Do not present constructed values as your own measurements.

E1. What changes, and what is measured in this comparison?

E2. Name two important controls and explain one.

E3. Recommend a product using the exact brief and a numerical result.

Optional actual product:	Dry / g	Wet / g	Retained / g
Trial 1			
Trial 2			
Trial 3			

E4. Record a test problem or one useful improvement. How would it improve the evidence?

Reflection. One choice I can now defend with evidence is …